

- Draw shapes.

In the next chapter, we'll continue the discussion of pygame and you'll see how to:

- Animate many objects.
- Build and react to a button.
- Create a text display field.

Bringing Up a Blank Window

As I said earlier, pygame programs run constantly in a loop, checking for events. It might help to think of your program as an animation, where each pass through the main loop is one frame. The user may click on something during any frame, and your program must not only respond to that input but also keep track of everything it needs to draw in the window. For instance, in one example program later in this chapter, we'll move a ball across the window so in each frame the ball is drawn in a slightly different position.

[Listing 5-1](#) is a generic template that you can use as a starting point for all your pygame programs. This program opens a window and paints the entire contents black. The only thing the user can do is click the close button to quit the program.

File: PygameDemo0_WindowOnly/PygameWindowOnly.py

```
# pygame demo 0 - window only

# 1 - Import packages
import pygame
from pygame.locals import *
import sys

# 2 - Define constants
BLACK = (0, 0, 0)
WINDOW_WIDTH = 640
WINDOW_HEIGHT = 480
FRAMES_PER_SECOND = 30
```